LYLE OKOTH

MACHINE LEARNING ENGINEER - SPEECH AND AUDIO PRODUCTION

Nairobi, Kenya | +254 797 668 066 | lyleokoth.mlengineer@gmail.com

linkedin.com/in/lyle-okoth | github.com/lyle-mlengineer | lyle-mlengineer.github.io | medium.com/@lyle-okoth

PROFESSIONAL SUMMARY

Machine Learning Engineer with **6+ years** taking speech models from checkpoint to production service: data pipelines, training, inference APIs, evaluation gates and cost control. End to end on Python, PyTorch and Hugging Face stacks served through FastAPI, Celery, Docker and AWS.

Served **8,000 requests/day at 180ms p95**, ran **50,000 hours of audio** through a diarization pipeline at **12% DER**, cut **training spend by 40%**, and shipped a regression harness that caught **100% of quality drops before release across 6 products**.

TECHNICAL SKILLS

- **Deployment and MLOps:** FastAPI, ONNX Runtime, TorchScript, WebSocket and SSE streaming, GPU batching, Docker, Kubernetes, GitHub Actions CI/CD, MLflow, Weights & Biases, rollback and drift monitoring
- **Model training:** PyTorch, Hugging Face Transformers, Whisper, VITS, LoRA / PEFT, quantization, distillation, mixed precision, DeepSpeed, Accelerate, CUDA, preemptible GPU capacity
- **Speech and audio:** TTS, ASR, speech translation, speaker diarization, audio feature extraction, librosa, FFmpeg
- **Evaluation:** WER / CER / MOS / DER, latency and p95 budgets, regression harnesses, gold sets, A/B testing
- **Data and infrastructure:** Python, SQL, PostgreSQL, Redis, Celery, Apache Kafka, Apache Airflow, DuckDB, Parquet, Elasticsearch, S3, Cloudflare R2, Azure

WORK EXPERIENCE

Machine Learning Researcher | SautiFlow Labs — 2026 - Present

Diffusion and flow-matching speech synthesis for languages with almost no public data.

- Trained a flow-matching TTS model covering **8 Kenyan languages** at **0.2s real-time factor** on a single consumer GPU, bringing multilingual synthesis inside interactive latency budgets.
- Cut **character error rate from 21% to 9%** on in-house Swahili and Sheng evaluation sets through fine-tuning, corpus curation and accent-aware text normalisation.
- Built a **120-hour** transcribed and force-aligned speech corpus from raw field recordings - the training base for the lab's synthesis and recognition models.
- Reduced **VRAM by 45%** with int8 quantization at a **0.4% accuracy cost**, cutting inference hardware per deployed voice.

AI Product Developer | Freelance — 2026 - Present

End-to-end ownership on a small number of products: research, flow design, model, launch, pricing, operation.

- Built **Voxa** alone from brief to live product: speech-skit generation, account system, **3 billing paths** and a studio interface for creators.
- Made infrastructure portable by design - storage swappable across local disk, **R2, S3 and Azure** with **no application code changes**.

Machine Learning Engineer | SautiFlow Labs — 2022 - 2026

Owned the path from checkpoint to service: pipelines, inference APIs, evaluation harnesses, cost control.

- Shipped the SemaScribe speaker-diarization pipeline, processing **50,000 hours of meeting audio at 12% diarization error rate**.
- Served speech inference on FastAPI + Celery + Redis at **8,000 requests/day at 180ms p95**, using ONNX Runtime, dynamic batching and GPU worker pools.

- Cut **training spend by 40%** with mixed precision and preemptible GPU capacity at **equal model quality**, verified on frozen evaluation sets.
- Automated ingestion of **200 hours of audio per week** with **98% label agreement across 3 reviewers** via ASR pre-labelling and reviewer rounds.
- Built a regression evaluation harness that caught **100% of quality drops before release across 6 shipped products**.
- Containerised deploys with Docker and GitHub Actions and tracked versions in MLflow for one-command rollback.

Software and Data Engineer | SautiFlow Labs **2022 - 2025**

Two sides of one job: move data reliably, and ship a product people can actually use.

- Built SnapSavanna, a dataset and labeling platform hosting **18 African-language datasets and 40,000 labeled samples** with content-hash versioning.
- Designed domain-driven FastAPI services split into **5 bounded contexts** with ports and adapters at the edges, keeping model code out of the product path.
- Shipped SuraYa and EchoBuni with Google OAuth, **M-Pesa and card billing**, and swappable object storage across **R2, S3 and Azure**.

SELECTED PROJECTS

SautiFlow - AI Audio Skit Studio

Script to finished skit: prompt expansion into speaker-labelled scenes, multilingual TTS per character, automatic timing, line-level re-takes, 15/30/60s export presets.

Tech: Python, FastAPI, PyTorch, VITS, Whisper, FFmpeg, Celery, Redis, PostgreSQL, Docker, AWS S3

EchoiID - Speaker Diarization and ASR

Who-spoke-when plus per-speaker transcription on stored files or live audio: voiceprint embeddings, clustering with overlap handling, word timestamps, SRT/VTT/JSON export.

Tech: PyTorch, pyannote.audio, Whisper, FastAPI, WebSockets, Redis, Celery, PostgreSQL, Docker, AWS

EDUCATION

BSc Computer Security and Forensics | Jaramogi Oginga Odinga University of Science and Technology (JOOUST) **[20XX - 20XX]**

ML, signal processing and security coursework; final-year project on audio watermarking for corpus provenance.

Mechatronics Engineering | Jomo Kenyatta University of Agriculture and Technology (JKUAT) **[20XX - 20XX]**

Fourier analysis, sampling theory, filter design and real-time embedded constraints - the grounding behind latency work.

OPEN SOURCE, WRITING AND COMMUNITY

- **Open ML releases:** 4 models and 3 datasets on Hugging Face, **12,000+ downloads**; model cards and training recipes per release.
- **Writing and tooling:** speech AI articles on Medium (@lyle-okoth); Python libraries for the YouTube and Google Drive APIs on PyPI.
- **Community:** OMDENA volunteer (2024); co-founder, SautiFlow Labs (2022).